

Repository & Guidelines Navigator

Quick Reference Map: Code Files vs. Guideline Instructions

1. Explicitly Mentioned Files & Folders

Grounded in Guidelines

`docs/`

[Documentation Folder]

Purpose Stores all project documentation, literature maps, and track instruction cards.

Action Read all included guides before building; update and log iterations in `RESULTS.md`.

`src/ & src/bsp/`

[Core Source Directory]

Purpose Houses the core Python codebase, 7-module pipeline harness, and track adapters.

Action Write your custom filtering and feature engineering code inside the track scripts here.

`docs/START_HERE.md`

[Core Guidance Documentation]

Purpose Provides the overall capstone roadmap, 5-step learning path, and pipeline integrity rules.

Action Read first to understand grading criteria, team vs. individual deliverables, and cross-validation rules.

`docs/BACKGROUND_MAP.md`

[Reference Documentation]

Purpose Maps textbook sections and domain literature review anchors to specific signal tracks.

Action Consult to find textbook refreshers and identify 5–8 academic sources to motivate design.

`docs/RESULTS.md`

[Core Deliverable / Graded Documentation (3 pts)]

Purpose Mandatory log for tracking experimental iterations, design choices, and labor division.

Action Update continuously with a new row for every test run, recording primary metrics and git hashes.

`docs/emg_ninapro_card.md`

[Reference Documentation]

Purpose	Outlines Ninapro DB1 dataset specifications, signal characteristics, and performance yardsticks.
Action	Read to understand DB1 properties and benchmark targets (~ 0.78 within-subject vs. ~ 0.12 new-subject).

`docs/emg_ninapro_instructions.md`

[Core Track Guidance Documentation]

Purpose	Track-specific requirements, design decision menus, and step-by-step guidance for surface EMG.
Action	Follow step-by-step to implement signal processing, feature construction, and evaluation modes.

`src/bsp/emg_ninapro.py`

[CORE TASK FILE TO EDIT]

Purpose	Main track execution script and entry point for running baseline and custom EMG experiments.
Action	Implement your custom signal processing inside <code>preprocess()</code> and feature engineering in <code>extract_features()</code> .

`src/bsp/adapter.py`

[Helper / Support Infrastructure Script]

Purpose	Defines the 7-module track harness (<code>EMGNinaproTrack</code>), default classifiers, and Ch. 8 noise-filtering functions.
Action	Import helper functions (e.g., <code>denoise</code> , <code>bandpass_notch</code>) into your track script; do not modify harness mechanics.

`src/bsp/report.py`

[Helper / Output Support Script]

Purpose	Generates formatted evaluation summaries, confusion matrices, and primary metric outputs with spread.
Action	None (called automatically by harness); copy its printed <code>rep["summary"]</code> outputs directly into <code>RESULTS.md</code> .

2. Unmentioned Files & Folders

Inferred Best-Guess Roles

NinaPro_Mat/

[NOT MENTIONED IN GUIDELINES]

Category Data Storage Directory

Best Guess Local directory containing raw MATLAB (.mat) signal files downloaded from the Ninapro DB1 dataset.

data/raw/

[NOT MENTIONED IN GUIDELINES]

Category Data Storage Directory

Best Guess Standard target folder where raw dataset records or offline synthetic fallback files are stored.

run_baseline.py

[NOT MENTIONED IN GUIDELINES]

Category Root Python Shortcut

Best Guess A root-level shortcut script to trigger `src/bsp/emg_ninapro.py` without typing long subfolder paths.

visualize_emg.py

[NOT MENTIONED IN GUIDELINES]

Category Plotting Utility

Best Guess Utility for plotting raw, filtered, or windowed EMG waveforms and channel spectra during exploratory data analysis.

requirements.txt

[NOT MENTIONED IN GUIDELINES]

Category Environment Config

Best Guess Defines exact Python dependencies (NumPy, SciPy, scikit-learn) required for reproducible execution.

.gitignore & .gitattributes

[NOT MENTIONED IN GUIDELINES]

Category Git Configuration

Best Guess Excludes large binary datasets and cache files from Git, and standardizes line endings.